

Project Work Natural Language Processing Summer 2026

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Start: 15.06.2026 12:00 (via Moodle)

Deadline: 25.06.2026 12:00 (via Moodle)

Grading Criteria

General

- Your solution solves the task and has sufficient quality.
- Your solution is well founded and well justified. Explain your solution in the documentation.
- Do not limit yourself just to techniques from the lecture but also research other possible approaches to find the best way to solve the project. Include current knowledge in the field and the current literature
- Your solution is efficient and effective (do the right things, do the things right).
- Your solution exceeds the quality obtained by AI tools when they are asked to solve the task.
- Your solution demonstrates a deep understanding of the problem.

Code

- Code must be written in Python (except otherwise specified)
- Your code is well structured (packages, classes, methods, ...), easy to read, understandable, and there are sufficient comments in the code. Uncommented code will be down-graded.
- In addition to comments in the code, every function must contain an appropriate docstring. You can follow the NumPy docstring guide: <https://numpydoc.readthedocs.io/en/latest/format.html>. Notice, a Sphinx documentation is not required.
- As a rule of thumb: The more complex the function, the more comments are required in the code.
- Your code is efficient, understandable, and written in a way that is not error-prone.
- Wherever possible, use available Python packages. Restrictions might be specified in the project description.
- Your code must run on the computers in the GPU lab (DC 1.07).

Documentation

- Your documentation must be a PDF. The use of $\LaTeX$ is recommended, but not obligatory. However, the documentation shall be in a proper report format (see grading). Markdown files or Jupyter notebooks hardly fulfill good report criteria.
- Your documentation presents your solution. Avoid unnecessary information in the documentation.
- It must be written in a way that another AI master student, who is not an expert in the field of the project task, could follow what you did and why you did it.
- It must be well-structured and written in proper language.
- Tables and figures shall be on point, clear, and concise.
- Each step in your solution must be well justified in the documentation.
- List all your references, use a proper scientific citation standard.

Project – Equations Knowledge Graph

Task Description

Your goal is to design a system to extract information from scientific papers. As in winter, it is about quantum physics papers.

Your solution now shall extract the following information from arbitrary quantum physics research papers:

- Any enumerated equation in the paper. Note: Physics papers usually contain lots of equations, only the important ones are enumerated, usually noted by a number in brackets (e.g. "(1)")
- Key information about the meaning of the equation
- Meaning of all abbreviations in the equation
- Relations between the equations: define meaningful relationships between each pair of equation, store it in a graph-ready format as outlined below.

In addition, you shall show, how the extraction works by adding an "audit trail". You find details on the dataset requirements in the following section.

Requirements

Your methods shall be generally valid for arbitrary quantum physics papers. For this project you build a prototype with 350 to 356 equations. For each exam ID, you find a paper list in Moodle ("paper_list_<exam ID>.txt"). Please work only with the list assigned to your ID and the papers in exactly that order. Violating this requirement leads to a fail.

For your prototype dataset, it is sufficient to use the *first 7 relevant equations per paper*. If the paper contains less than 7 equations, please use all, otherwise cut off any further equations. Stop your dataset creation as soon as it contains 350 equations, but the last paper needs to be processed *completely*. That means, the cut-off is not necessarily done at exactly 350 equations. As long as your dataset has less than 350 equations, you analyze a further paper.

You shall use the papers from the arxiv.org platform. **When automatically querying information from servers, you are responsible for respecting the robots.txt and for using appropriate sleep times between requests for avoid server overloading.**

The dataset shall be a JSON file with the arxiv ID of the paper as main key. Below the main key, there is an equation dictionary, which has a key for each equation. Every paper must have a key in the dataset. In case it does not contain any enumerated equations, the enumeration dictionary shall be empty. The equation key shall be the equation number as stated in the respective paper. Behind the equation key, there is a dictionary with the relevant information per equation, which must have the following structure.

- equation: The equation in latex code
- meaning: A short but meaningful description what the equation expresses, if applicable, what is the name of the equation (e.g. "wave function").
- symbols: A dictionary explaining all relevant abbreviations in the equation (see below).
- relations: a dictionary of relations to all other relevant equations in the paper (see below).

- audit-trail: an audit trail of the methods involved in extracting the information (see below).

The symbols dictionary has the following sub-structure: every symbol is a key, the value is the corresponding description, e.g.

```
{
  i: imaginary unit,
  phi: wave function
}
```

For symbols, use the latex command without the backslash (e.g. "phi" for ϕ). Mathematical standard operators (+, -, ∇ , ...) are not required to be explained in the symbols dict.

The dictionary of relations contains every other equation key for that paper and describes the relation to that equation as a dictionary with keys "grade" and "description". The grade shall classify the relation into the three categories "none" for no relation, "strong" for a clear relation, "potential" in case it is not fully clear but some relation might exist. A concept for this classification shall be derived and presented in the documentation. In case the grade is "strong", there must be a description of the relation with a few words (e.g. "equivalent", "special case", "negation", etc.), in case it is "potential" there should be a description.

Example:

```
{
  1: {
    equation: a + \phi > 0
    meaning: Albert-Weil formulation of quantum probability,
    symbols: { ... }
    relations: {
      2: {
        grade: strong
        description: is a special case
      }
    },
    audit-trail: {...}
  },
  2: {
    equation: a + \phi = 0
    meaning: Albert-Weil formulation of quantum probability at zero temperature,
    symbols: { ... }}
    relations: {
      1: {
        grade: strong,
        description: is the general formulation,
      }
    },
    audit-trail: {...}
  }
}
```

The audit-trail must be a dictionary with the following requirements: The keys are the methods you implement in your code that are used to extract information. The values is a short but meaningful output about

the extracted information by that method. To keep the audit trail meaningful and usable, the value must be precise and short.

Example: You extract an equation with the method `extract_eq()`, find symbols with the method `find_symbol()` and use the method `extract_symbol_name()` to obtain the specific name of that symbol. Then the corresponding part of your audit trail could (!) look like that:

```
{
  method extract\_eq: 'Found equation a+phi=0',
  find_symbol: 'found a',
  find_symbol: 'found phi',
  extract_symbol_name: 'extracted text '$a$ is a normalization constant'
  --> a: normalization constant',
  extract_symbol_name: 'extracted text 'the wave function $\phi$ is derived from
  a general argument' --> phi: wave function',
  ...
}
```

The audit trail must be suitable to analyze and justify the quality of your approach and your methods, to analyze its success or failure, and to prove compliance with the following method requirements:

- All information extraction must be done from arxiv (PDF, $\LaTeX$, or HTML) only.
- No prompting is allowed. Neither external APIs (like ChatGPT, Perplexity, Claude, etc.) nor language models hosted by yourself may be used with prompting (text generation).

Documentation focus

In your documentation, you must focus your discussion on the following aspects:

- Key ideas to solve the problem. Do not describe your code here, but what are the important solution ideas of your approach for all required data items and why did you choose them.
- Generalization of your approach: How well do you think your approach generalizes to arbitrary quantum physics papers and why? Present evidence.
- Quality of your approach: Discuss the quality of your results critically, present evidence. In particular, discuss aspects that could not be further improved and why.

Learning and Project-Specific Criteria

Learnings from the project: Prototyping an NLP application following customer requirements, development and evaluation of NLP approaches, information retrieval and processing, quality assurance and result analysis, presentation in structured, concise form, focus on key aspects.

- Deliverables must be complete and follow specifications and quality criteria above.
- If things are not specified, clearly state in the documentation which decisions you met instead.
- Your submitted dataset must have reasonable quality.
- Your approach must be sound, follow the rules, and generalize beyond the papers analyzed by you.

- Analyze success and error cases thoroughly, remember the discussions for the case studies in the lectures.
- The audit trail must be suitable for quality assurance and demonstrate that the rules have been met. Missing, incomplete, or unclear audit trails lead to downgrading.
- Your documentation must provide argumentation for your approach for a reader with general knowledge in AI (other master student) to understand what you did and why you did it.
- Documentation must obey scientific standards (regarding structure, language, citation, AI tool use declaration, ...).
- Documentation text (pure text without references, captions, tables, ...) must not exceed 1200 words.
- Documentation formatting: Arial 11 pt, one column, single line spacing. In $\LaTeX$, please use an article document class. Do not use any paper templates.

The following deliverables for this project must be part of your submission:

- The dataset in json format
- Your documentation as PDF.
- Your source code in a separate folder and zipped (zip format only, no rar, tar or else!).